MAIS PAC

IPhO 03.01: Measurement, Apparatus and Accuracy · Worked Solutions

Measurement, Apparatus and Accuracy: Worked Solutions

MAIS Physics & Astronomy Club (PAC) · Reasoning, derivations and checks

10 worked solutions · independent explanations · consistency checks

Make a genuine attempt before reading. Each solution explains how to choose the method, carries units through the calculation, and checks the result. The exercises are original MAIS PAC teaching problems, not official IPhO questions or marking schemes. Parameters are supplied locally; an exam's constants take precedence.

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Solution 1 — Reading a vernier. Return →

(a) 10 vernier divisions cover 9 {mm}, so the least count is 10-9 = 1 {mm} divided by 10, i.e. 0.1 {mm}.

(b) Raw = 23 + 6(0.1) = 23.6 {mm}.

(c) Corrected = 23.6 - 0.20 = 23.4 {mm}, with an uncertainty of about 0.05 {mm} (half the least count) plus the 0.05 {mm} uncertainty in the zero correction: (23.40 ± 0.10) {mm}. **Check:** the correction is small compared with the reading (0.2 in 23.4), but it would be 17% of a 1.2 {mm} reading.

Solution 2 — A micrometer reading. Return →

(a) Reading = 3.50 + 27(0.01) = 3.77 {mm}.

(b) Resolution = 0.01 {mm} (one thimble division), so the uncertainty is about ±0.005 {mm} to ±0.01 {mm}.

(c) At 3.77 {mm}: 0.005/3.77 = 0.13%. At 0.12 {mm}: 0.005/0.12 = 4% — thirty times worse. **Check:** this is the general rule: an absolute uncertainty becomes a large **relative** one for small readings, which is why measuring a thin wire or a short time is hard.

Solution 3 — Random or systematic? Return →

(a) **Systematic** (always the same direction); start the watch consistently or use a light gate.

(b) **Random** (drifts both ways over the experiment); repeat, plot the temperature, or use a water bath.

(c) **Systematic** (a zero error); tare the balance before use.

(d) **Random** (the eye's judgement varies); use a magnifier, take many readings, or use a digital sensor.

(e) **Systematic and random together**: the average diameter may be off systematically if the wire is tapered, and the readings scatter randomly about it. Measure at several points along the wire and quote the spread. **Check:** only the systematic ones fail to improve with repetition, which is the operational test for the distinction.

Solution 4 — Propagation through a formula. Return →

(a) g = (4π²(0.994))/((2.00)²) = 9.81 {m,s^(-2)}.

(b) Fractional uncertainties: T contributes 2×(0.02/2.00) = 2.0%, L contributes 0.002/0.994 = 0.2%. Adding in quadrature, Δ g/g = √(2.0²+0.2²)% ≈ 2.0%, so g = (9.81 ± 0.20) {m,s^(-2)}.

(c) The period dominates because it is squared and because a manual stopwatch is imprecise. Timing many oscillations (Problem 9) is the standard fix. **Check:** a 2% uncertainty is far too coarse to distinguish 9.81 from 9.78, so the experiment as described cannot measure g well — which is exactly the sort of thing an examiner wants you to notice.

Solution 5 — Averaging. Return →

(a) σ_x = s/√()N = 0.12/√(10) = 0.038 {mm}, so the result is (52.30 ± 0.04) {mm}.

(b) To get 0.02 {mm}: √()N = 0.12/0.02 = 6, so N = 36 readings.

(c) Beyond about 10–20 readings a systematic effect (the ruler's calibration, the object's non-uniformity, drift) usually exceeds the statistical uncertainty, so the 1/√()N law stops working. **Check:** 36 readings for a factor of 2 improvement is a poor return, which is why better instruments beat more readings.

Solution 6 — Calipers versus ruler. Return →

(a) $A = \pi d^2/4 = \pi(1.84\times10^{-3})^2/4 = 2.66\times10^{-6}$ {m²}. The diameter has 0.01/1.84 = 0.54% uncertainty, doubled to 1.1% for the area: $A = (2.66 \pm 0.03)\times10^{-6}$ {m²}.

(b) With $d = 2 \pm 1$ {mm}, $A = (3 \pm 3)\times10^{-6}$ {m²} — a 100% uncertainty, which is no measurement at all.

(c) Resistivity is $\rho = RA/L$, so the diameter enters squared (through A); a 50% error in d becomes a 100% error in ρ. **Check:** the area ratio between the two methods is about 1.5, so even the **value** disagrees, not just the precision — a ruler cannot measure a 2 {mm} wire.

Solution 7 — Significant figures. Return →

(a) Mass: 0.1/54.3 = 0.18%; volume: 0.2/20.0 = 1.0%; total ≈ 1.0% (the volume dominates).

(b) $\rho = 2.71\times10^3$ {kg,m^(-3)} with ± 1%, i.e. $(2.71 \pm 0.03)\times10^3$ {kg,m^(-3)}.

(c) The six digits claim a precision of one part in a million, while the data justify about one part in a hundred. Quoting unjustified digits is not a harmless habit: it misrepresents the quality of the experiment. **Check:** the rounded value, 2.7×10^3, is close to aluminium (2700), which is the kind of consistency check that reassures you the arithmetic is right.

Solution 8 — A systematic error hunt. Return →

(a) Mean = 10.18 Ω; the spread is about ±0.1 Ω, so the standard uncertainty of the mean is roughly $0.1/\sqrt{5} = 0.04$ Ω: $R = (10.18 \pm 0.05)$ Ω.

(b) The stated value is 9.8 ± 0.1 Ω. The difference, 0.38 Ω, is about four times the combined uncertainty, so it is significant — there is a systematic problem, not bad luck.

(c) Plausible causes: (i) the ammeter has a resistance, so the measured current is slightly low — use a voltmeter across the resistor only, or correct for the ammeter; (ii) the voltmeter draws current, so the measured current is slightly high — repeat with a high-impedance meter; (iii) the resistor heats up and its resistance drifts — keep the current low and check for drift. A fourth possibility is a zero offset in either meter, testable by short-circuiting the ammeter and comparing the voltmeter with a known source. **Check:** every one of these produces a **consistent** offset, which is exactly why more repeats would never have revealed it.

Solution 9 — Timing with a stopwatch. Return →

(a) Mean time for 20 oscillations = 40.2 {s}; the reaction-time uncertainty is about 0.2 {s} for a single start-stop pair (and about 0.2 {s} in total, since both the start and the stop are uncertain). The period is 40.2/20 = 2.01 {s} with uncertainty 0.2/20 = 0.01 {s}: T = (2.01 ± 0.01) {s}.

(b) Timing one oscillation would give 2.0 ± 0.2 {s} — a 10% uncertainty. Timing N oscillations divides the absolute uncertainty by N, so a 20-swing timing reduces it to 1%.

(c) $g = 4\pi^2 L/T^2$, so $\Delta g/g = 2\Delta T/T = 2(0.01/2.01) = 1.0\%$: g = (9.7 ± 0.1) {m,s^(-2)} for L = 1.00 {m}. **Check:** timing more oscillations (say 50) is the cheapest improvement, but beyond a few hundred the pendulum's amplitude decays and the frequency drifts.

Solution 10 — Designing a measurement. Return →

(a) Mass: a digital balance reading to 0.01 {g} — the uncertainty is 0.01/30 = 0.03%, negligible. Volume: the stone is irregular, so measure it by displacement in a narrow measuring cylinder or, better, by weighing it in water (Archimedes). A 25 {mL} cylinder reads to 0.5 {mL}, giving 0.5/12 = 4% for a 12 {cm³} stone — too coarse for 1%; a 10 {mL} burette with 0.05 {mL} divisions gives 0.4%.

(b) Balance: 0.03%; burette displacement: about 0.4%; so the combined uncertainty is about 0.4%, comfortably inside the 1% target.

(c) The volume limits the accuracy, because the stone is small and irregular. Improve it by using a larger sample, by using a burette rather than a cylinder, by coating a porous stone to stop water soaking in, and by repeating the displacement several times. **Check:** if the stone is porous, the volume measured is the envelope plus the water absorbed — a systematic error that no amount of repetition removes, which is why the coating matters.